Analysis of Coffee Shop

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Agenda

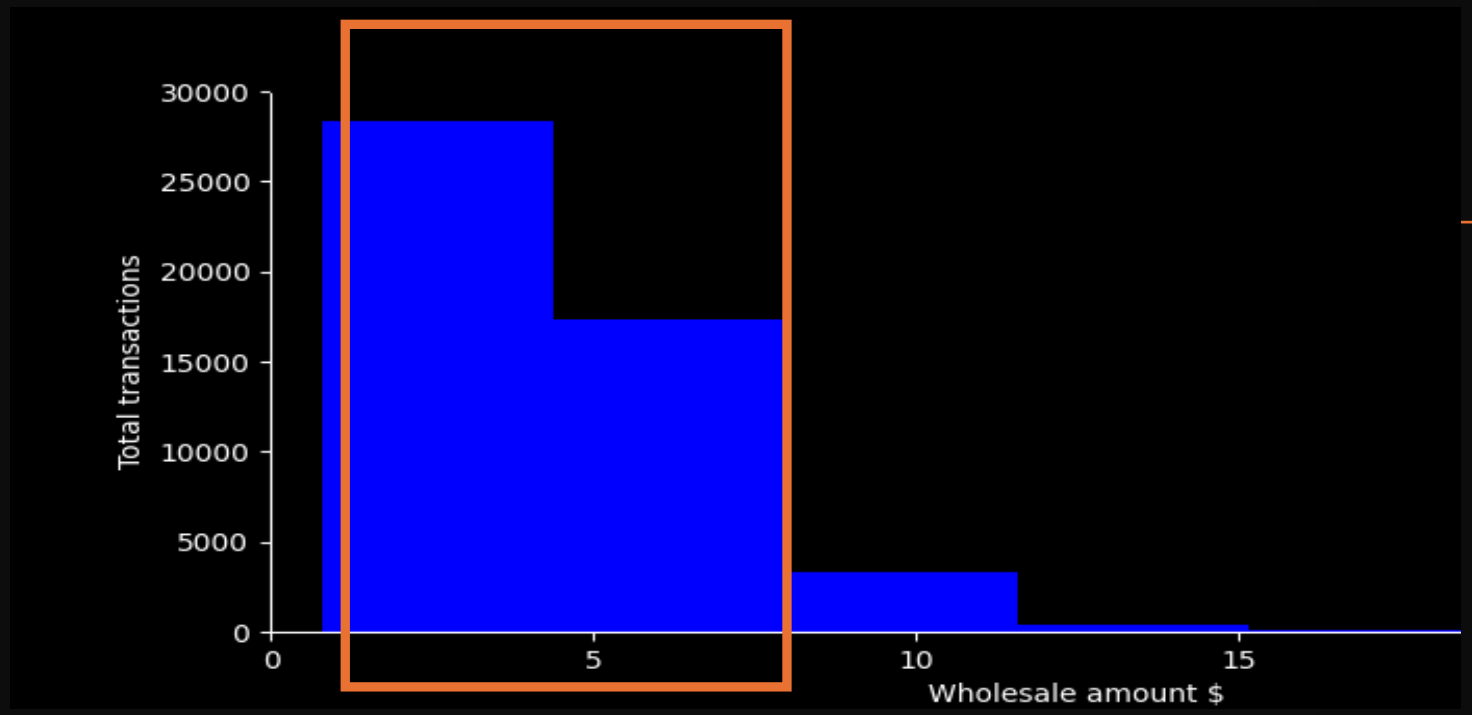
- Key factors influencing coffee shop sale
- Customer generations and weekday sales patterns
- Distribution of Transactions
- Product types and associated profits
- Correlations among numeric factors
- Key Insights and actionable recommendations

Total transactions by weekdays

Weekday	Total Transactions
Monday	8471
Wednesday	7105
Tuesday	6978
Friday	6937
Saturday	6926
Thursday	6831
Sunday	6646

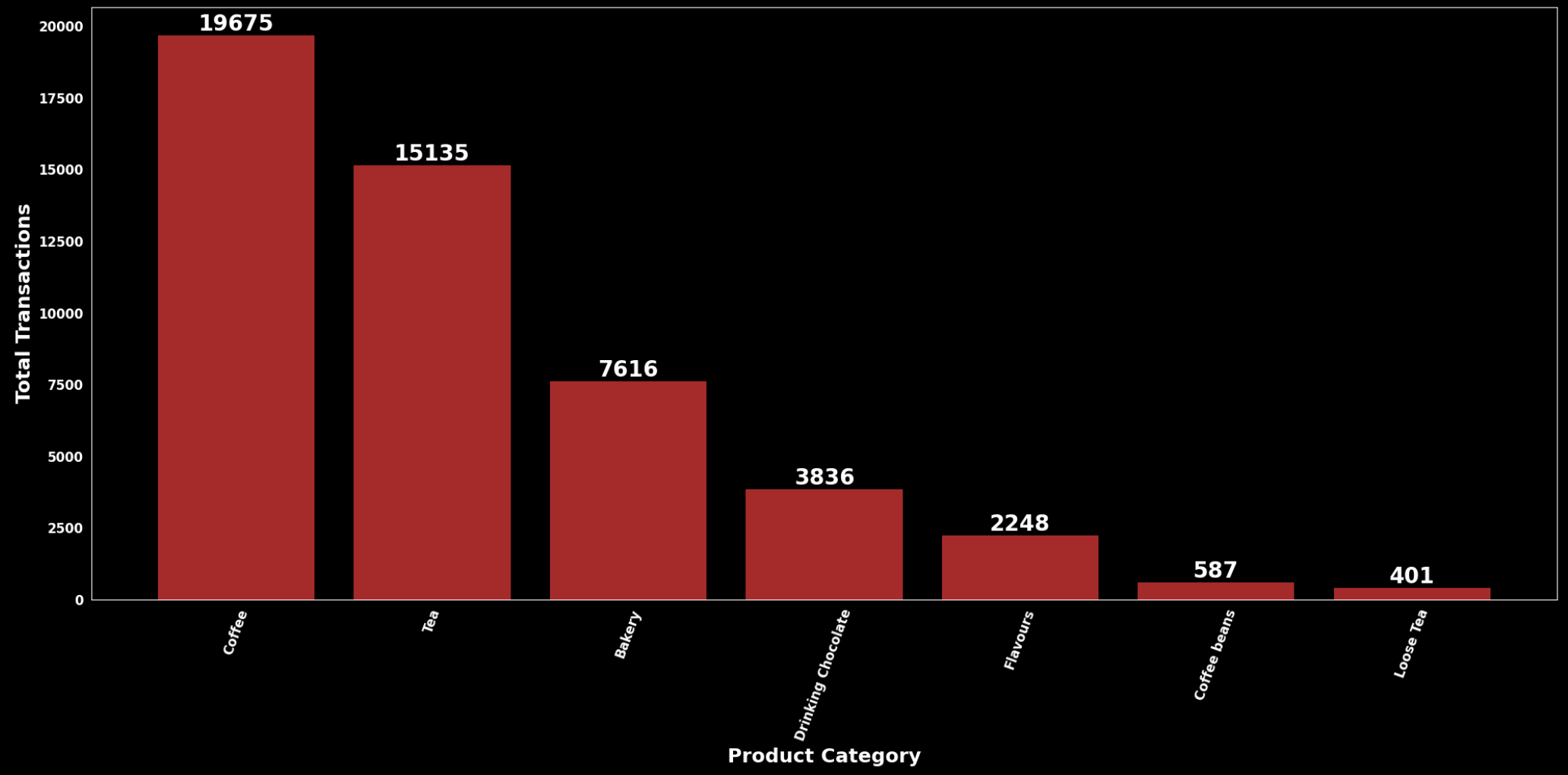
- Most transactions occur on Monday, while Sunday has the fewest.
- To encourage more customer activity, it is recommended to introduce special offers or promotions on Sundays.

Distribution of Transaction Amount

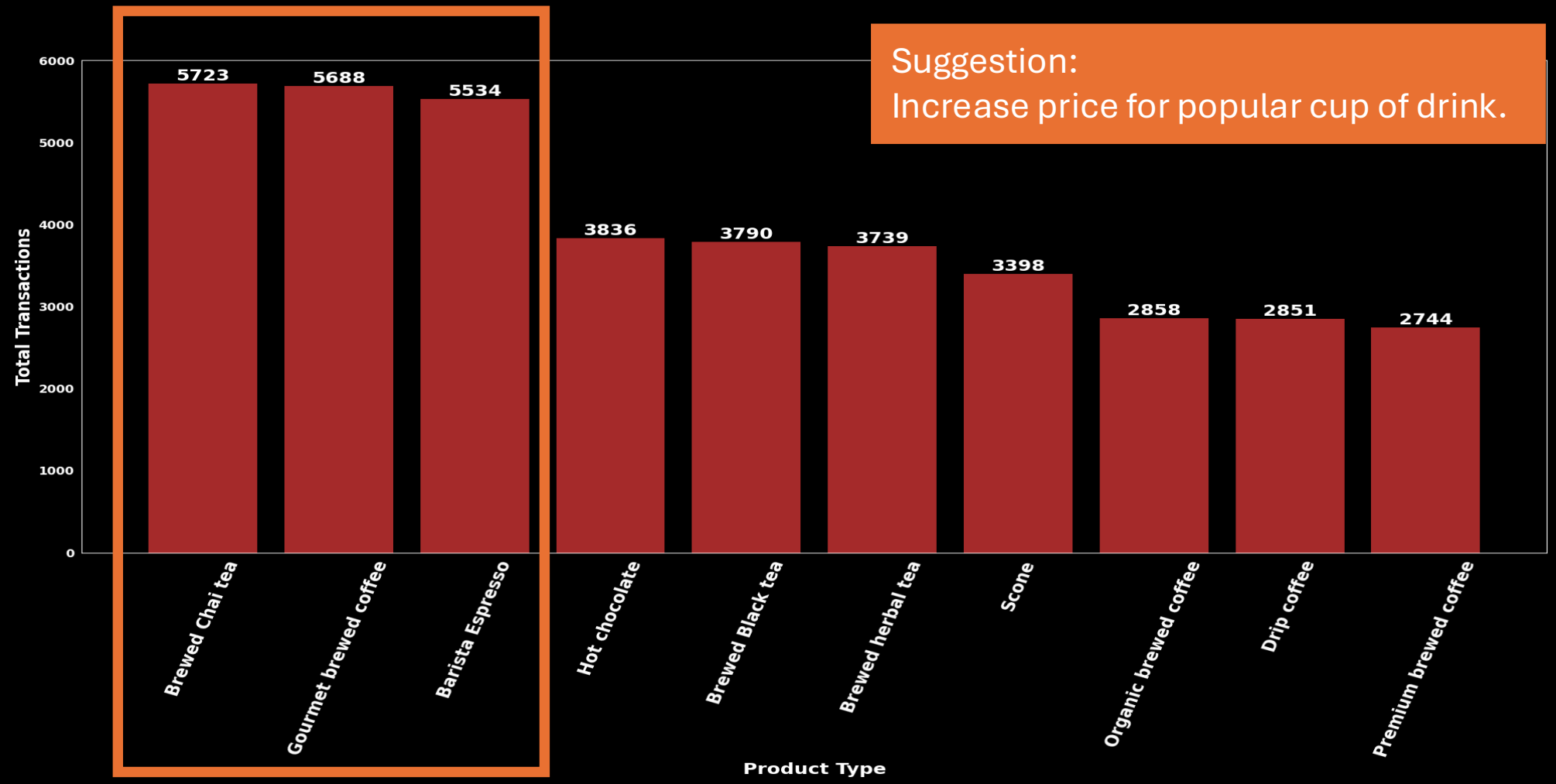


Most of the transactions between **$2 and $8,** indicating that customers mostly purchase a single cup of drink. To increase the value of each transaction, it is recommended to offer customers additional items, such as syrups or pastries.

The Most Popular Product Category



The Most Popular Product Type



Suggestion:
Increase price for popular cup of drink.

Median profit group by product type

Highest Profit

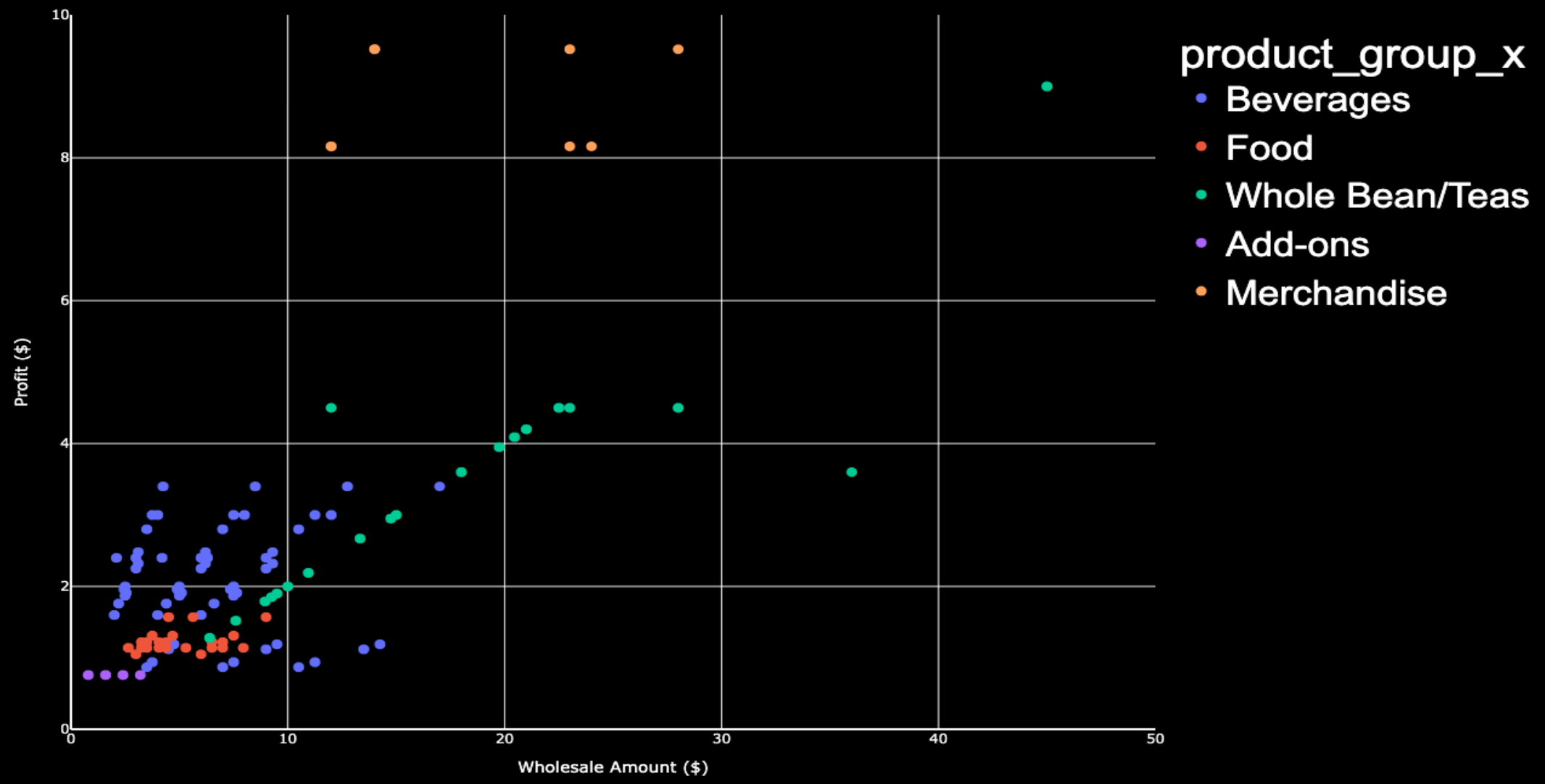
Product type	*Profit $*
Clothing	19.04
Housewares	8.95
Premium Beans	6.82
Organic Beans	4.04
Gourmet Beans	3.70
House Beans	3.60
Espresso Beans	3.48

Medium Profit

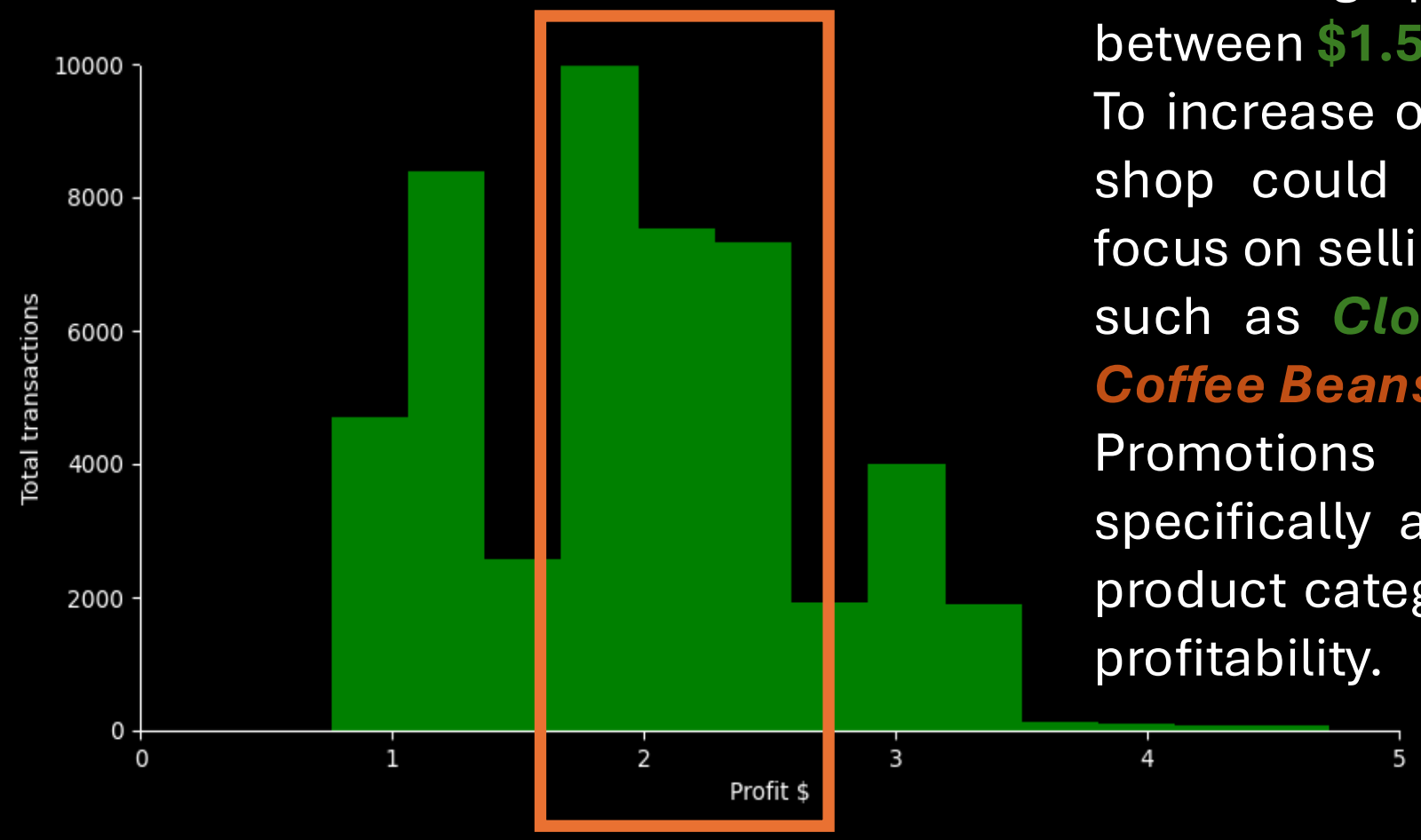
Product type	*Profit $*
Barista Espresso	2.96
Premium brewed coffee	2.50
Organic brewed coffee	2.31
Brewed Chai tea	2.20
Gourmet brewed coffee	2.16
Drinking Chocolate	2.09
Brewed tea	2.06
Green beans	2.00

The products as *Pastry, Hot Chocolate, Syrups* have a **smallest profit.**

Profit & Total Wholesale Amount



Distribution of Profit



The average profit per transaction is between **$1.5 and $2.5.**

To increase overall profit, the coffee shop could either raise prices or focus on selling higher-margin items, such as ***Closing, Housewares,*** *or* ***Coffee Beans.***

Promotions could be targeted specifically at the ***"Coffee Beans"*** product category to boost sales and profitability.

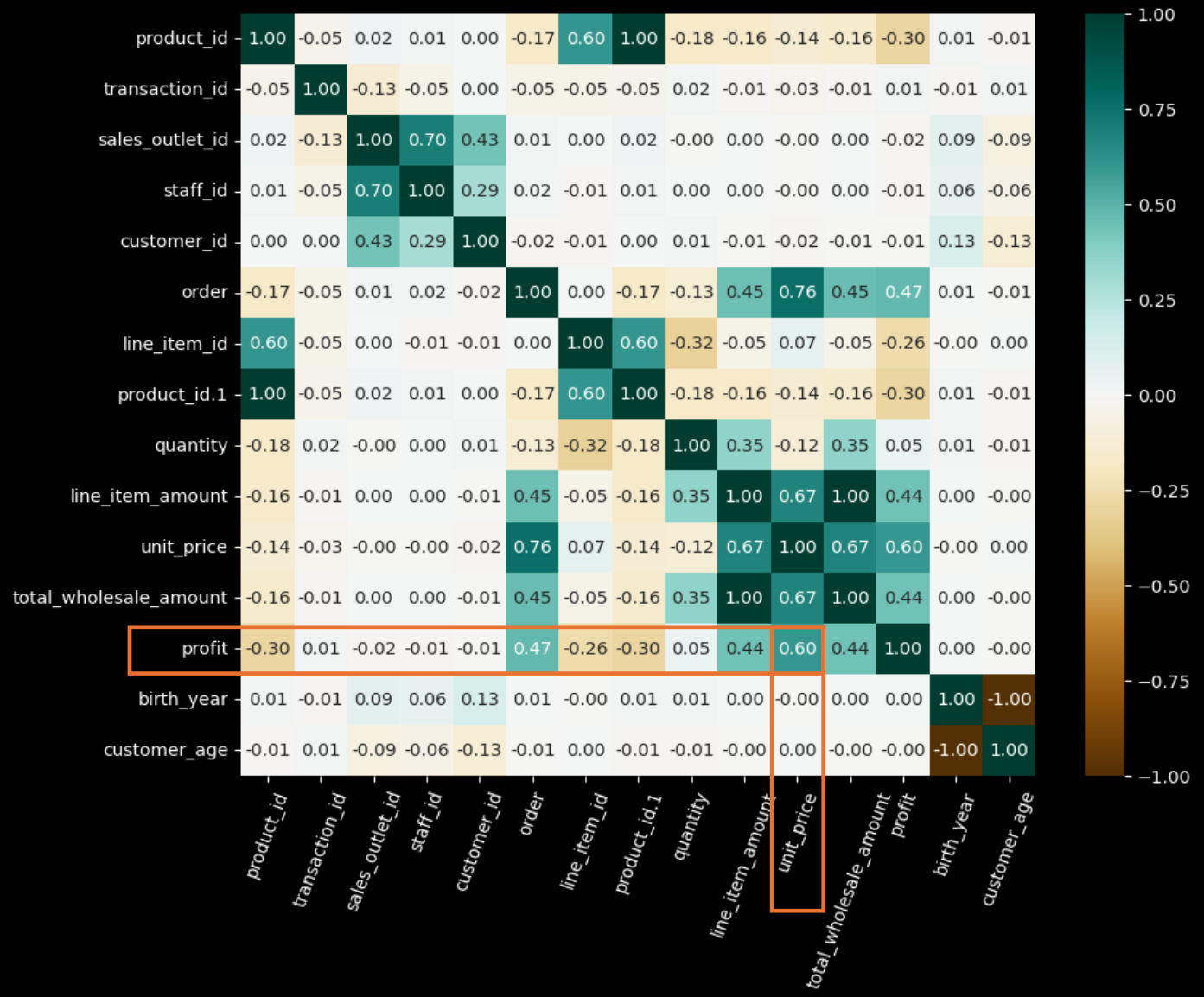
Total Transactions by Generations

More than half of the transactions **do not** include customer ***generation*** information because many customers did not register their details.

Due to this data gap, we **can not** confidently conclude that the ***Baby Boomer*** generation is more loyal or makes more purchases.

Generation	*Total Transactions*
Baby Boomers	**6019**
Gen X	**5691**
Older Millennials	**5474**
Gen Z	**4289**
Younger Millennials	**3379**

Correlation between numeric values



The strongest correlation is observed between **Unit Price and Profit** indicating that higher-priced items tend to generate higher profits (correlation ≈ 0.60)

Additionally, **Total Wholesale Amount** shows a strong relationship with both **Profit** and **Unit Price** (correlation ≈ 0.67)

Key Insights:

- Sales are consistent across weekdays with minimal variation
- Most transactions generate only a small profit
- High-profit items are sold infrequently
- Profit is strongly linked to the total wholesale amount

Recommendations

- **Promote high-profit items** with bundles or special offers.
- **Engage customers** through loyalty programs and personalized deals.
- **Collect new customer data** to tailor marketing and improve retention.
- **Optimize product mix** to focus on revenue-driving items.
- **Use weekday promotions** to increase sales during slower periods.

References

- Kaggle dataset https://www.kaggle.com/code/ahmedabbas757/coffee-shop-sales
- Jupiter Notebook
- Python Visualization